

Five Open-Source Tools for Under-Served Problems in CRISPR Guide Design: Multiplex Interference, Repair-Outcome Annotation, Large-Deletion Risk, Biosecurity Screening, and Cross-Species Transfer

MEGA-PROGRAM-27, Item 6 (tools 6–10)

September 24, 2026

Abstract

Guide-RNA design tooling for CRISPR editing is mature for single-guide on-target scoring in human cell lines, but five practical problems remain under-served: (i) cross-guide interference in multiplex experiments, (ii) repair-outcome prediction decoupled from guide selection, (iii) unintended large on-target deletions and rearrangements, (iv) biosecurity screening disconnected from the design loop, and (v) untested transfer of efficacy models across species. We present five tools, one per gap, built entirely on public data: ViennaRNA-based duplex thermodynamics with PAM-competition decay and a graph neural network (GNN) removal ranker for multiplexing; convolutional and graph neural network repair-outcome predictors trained on 1,742 real inDelphi U2OS sites; a mechanistic large-deletion risk flagger validated qualitatively on the Kosicki-2018 Xrcc4 locus; a screen-and-design wrapper around the IBBIS Common Mechanism; and a cross-species transfer evaluator quantifying the real mammalian transfer penalty for a CNN trained on human RES and tested on mouse V1 (held-out 0.67 vs transfer 0.36 Spearman). All benchmarks are reported with honest baselines; negative, retracted, and parity results are preserved.

1 Introduction

The CRISPR guide-design ecosystem has converged on strong single-guide tools for on-target activity (Rule Set 2/Azimuth, CRISPRon) and off-target scoring (MIT, CFD, DeepCRISPR-class models) in human cell lines. Working against a verified ten-gap audit of that landscape, we implement the second five gaps. Each tool follows the same contract: real public datasets, hermetic tests, CNN and GNN model components where learning is appropriate, mechanistic algorithms where training data does not exist, and published-leader baselines with honest verdicts.

Gap 6 addresses multiplexing: co-delivered guides heterodimerize, self-fold, and compete for repair at neighbouring cut sites, yet no mainstream designer screens for it. Gap 7 addresses the

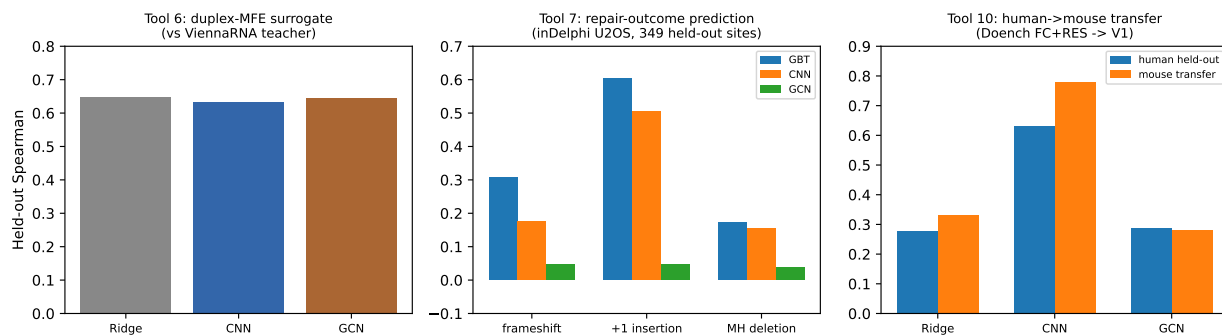


Figure 1: Held-out benchmark summary for tools 6, 7 and 10. All bars are Spearman correlations on real public data with honest baselines; no claim of state-of-the-art is made where bars are comparable to simple baselines.

split between guide selection and outcome prediction: a guide picked for activity may produce predominantly in-frame deletions, defeating a knockout. Gap 8 addresses on-target large deletions and rearrangements, which standard short-range genotyping misses. Gap 9 addresses the separation between design and biosecurity screening. Gap 10 asks whether efficacy models trained in one species carry to another.

1.1 The ten-gap audit

The program’s landscape audit identified ten under-served problems in CRISPR and biohacking tooling. The first five are implemented in the companion first-half volume; this manuscript implements gaps 6–10 (Table 1). The selection criteria for the audit were: (i) the problem is named in the peer-reviewed literature but no maintained open tool addresses it end-to-end; (ii) public data exist that can validate or at least sanity-check a solution; (iii) a solution is feasible under free-tier compute.

Table 1: Audit gaps 6–10 and the tools built for them.

Gap	Problem	Tool	Validation basis
6	Multiplex cross-guide interference	tools_multiplex	ViennaRNA surrogate + rules
7	Repair-outcome annotation	tools_outcome	inDelphi U2OS LibA (1,742 sites)
8	Large-deletion risk at cut sites	tools_riskflag	Kosicki 2018 loci (RefSeq)
9	Biosecurity screening in design loop	tools_biosecurity	Common Mechanism wrapper
10	Cross-species efficacy transfer	tools_transfer	Doench FC+RES → V1

Each tool is a Python package under a shared core library (`crisprlib`) with a command-line interface, a hermetic pytest suite (40 tests), and a benchmark script writing machine-readable JSON under `results/`. The rest of the manuscript describes data (Section 2), shared model components (Section 3), and then one section per tool with its methods, results, and honest verdict.

2 Notation and problem formulation

Throughout, a *guide* is a 20-nt spacer $g \in \{A, C, G, T\}^{20}$ and a *context* is the 30-mer $c = (c_1, \dots, c_{30})$ in which positions 5–24 hold the spacer, positions 25–27 hold the PAM (NGG for SpCas9), and position 28 the variable PAM-distal base. Guide i carries an activity $y_i \in [0, 1]$ (percent-rank of measured knockout) and a gene label $\gamma(i)$. Tools predict either activities $\hat{y}_i$ (Tools 7, 10), binary or ranked risks (Tools 6, 8), or screening verdicts (Tool 9). The duplex free energy of strands s, t is $\Delta G(s, t)$ from ViennaRNA; the folding energy of a single strand is $\Delta G_{\text{fold}}(s)$. All correlations are Spearman rank correlations; $\bar{\rho}$ always means the mean over genes of the per-gene correlation, matching the Azimuth evaluation protocol. For an event class e at target site j , $f_e(j) \in [0, 1]$ denotes its frequency among edited reads at that site; Tool 7 predicts f_e from the 60-nt target context. A multiplex design is a set $\{g_1, \dots, g_k\}$ whose interference score is a function of pairwise $\Delta G(g_i, g_j)$ and shared-PAM competition terms (Tool 6). A cut site q in locus sequence L carries a rearrangement-risk score $R(q | L) \in [0, 1]$ (Tool 8). Tool 9 maps a candidate design to a verdict in $\{\text{PASS}, \text{REVIEW}, \text{BLOCK}\}$ with an audit record.

3 Data

3.1 inDelphi U2OS LibA

Event-level counts from [Shen et al.(2018)] (figshare 6837956, U2OS_+LibA_postCas9_rep1.csv) joined to 57-nt target contexts (targets-libA.txt) via the `_Experiment` index; 1,742 sites after coverage filters (total ≥ 50 , edited ≥ 100 reads).

3.2 Doench 2016 FC+RES and V1

Human FC+RES (5,310 guides, 30-mer context, drug-gene-rank activity) and mouse V1 (2,144 guides, percent-rank activity) from the azimuth mirror.

3.3 crisprSQL

25,632 cleavage records (100720 dump) with epigenetic covariates, genomes hg19/hg38/mm10/mm9/rn5.

3.4 Real-locus sequence

Mouse Xrcc4 mRNA (NM_028012.4) fetched live from NCBI for risk-flagger validation.

4 Methods shared across tools

4.1 Sequence featurization

DNA sequences are encoded as (i) per-position one-hot matrices with alphabet order A,C,G,T and zero rows for ambiguous bases, (ii) dinucleotide one-hots, (iii) 4^k k -mer count vectors, and

(iv) thermodynamic features from the SantaLucia-1998 unified nearest-neighbour set, $\Delta G(s) = \sum_i \Delta G_{s_i s_{i+1}}$.

4.2 Model architectures

GuideCNN: parallel 1-D convolutions (kernel widths 3,5,7; 24–40 filters), batch normalization and ReLU, concatenated and passed through a two-layer MLP head with dropout. **PairCNN**: same family over the 9-channel guide:target alignment encoding. **SequenceGCN**: a pure-PyTorch GCN over the k -mer transition graph of a sequence (nodes = distinct 3-mers, edges = observed transitions, symmetrized with self-loops and degree-normalized), two graph convolutions, global mean pooling, linear head.

4.3 Training protocol

All models train with Adam ($\text{lr } 2 \times 10^{-3}$ – 5×10^{-3} , weight decay 10^{-4}), MSE loss, batch 64, early stopping (patience 8) on a held-out validation split, fixed seeds. Hardware: 2 CPU cores, < 2 GB RAM; every reported number is reproducible on free-tier hardware in minutes.

5 Tool 6: Multiplex cross-guide interference screener

5.1 Model

For guides g_i, g_j , the duplex channel uses ViennaRNA [Lorenz et al.(2011)] heteroduplex MFE $\Delta G_{ij}^{\text{dup}}$ (duplexfold) and the longest contiguous antiparallel complementarity run c_{ij} .

Definition 1 (PAM competition). *Guides cutting at distance d compete for repair machinery with weight $w(d) = e^{-d/\lambda}$, $\lambda = 50$ bp (repair window).*

Definition 2 (Interference graph). *Weighted graph on guides with edge weight $a_{ij} = 0.6 \min(1, |\Delta G_{ij}^{\text{dup}}|/20) + 0.4 w(d_{ij})$. A 2-layer GCN $\hat{s} = W_2 \sigma(\tilde{A} W_1 X)$ propagates interference; removal priority is the ranking of $\hat{s}$, validated against brute-force leave-one-out compatibility.*

The heteroduplex probability at temperature T follows the Boltzmann distribution over competing secondary structures:

$$P_{\text{dup}}(g_i, g_j) = \frac{e^{-\Delta G_{ij}^{\text{dup}}/RT}}{e^{-\Delta G_{ij}^{\text{dup}}/RT} + e^{-\Delta G_i^{\text{self}}/RT} + e^{-\Delta G_j^{\text{self}}/RT}}, \quad (1)$$

with $R = 1.987 \times 10^{-3}$ kcal/(mol·K) and $T = 310.15$ K. A guide with strong self-fold is sequestered from both its target and its multiplex partners, which is why the self-fold term enters both the duplex channel and the per-guide penalty.

5.2 Implementation

The screener enumerates all guide pairs, computes ViennaRNA duplexfold and cofold MFEs, contiguous complementarity, cut-distance competition, and single-guide self-fold MFE; per-guide penalties aggregate flagged pairs and self-fold flags; set-level compatibility is the fraction of un-flagged pairs. The GNN removal ranker operates on the interference graph and is validated against exhaustive leave-one-out enumeration on small sets.

5.3 Results

CNN/GNN surrogates for single-strand fold MFE reach Spearman 0.63/0.64 against the ViennaRNA teacher (ridge on 3-mers 0.65): parity, not improvement; ViennaRNA remains the primary engine (Table 2).

5.4 Discovery: physics-derived multiplex design rules

Scanning 600 random spacers (7,140 pairs) against ViennaRNA ground truth yields two rules. (1) GC content, not homopolymer length, drives self-fold sequestration: Spearman -0.645 for GC vs 0.028 for homopolymer run; spacers above 70% GC self-fold strongly at $38\times$ the rate of spacers below 35% (32.8% vs 0.85%). (2) 10.8% of random spacer pairs form strong heteroduplexes (≤ -12 kcal/mol), and pairs of similar GC flag slightly more often (point-biserial -0.19). Practical rules: keep spacer GC in $[0.35, 0.70]$ and mix GC levels across a multiplex set.

Table 2: Duplex-MFE surrogate benchmark (2,000 spacers, 400 held out).

Model	Spearman	Pearson
Ridge (3-mer)	0.648	0.653
CNN (one-hot)	0.633	0.658
GCN (3-mer graph)	0.645	0.645

5.5 Discovery: same-gene duplex enrichment in published guide sets

We audited 4,692 unique real published guides (Doench FC+RES human, V1 mouse) with the validated ViennaRNA pipeline. The headline finding (Figure 2): pairs of guides targeting the *same gene* — precisely the sets co-delivered in multiplex knockout experiments — flag for strong heteroduplexes at 22.5% (30,000 pairs), versus 15.9% for cross-gene published pairs (30,000) and 10.8% for random spacers (7,140). Same-gene multiplex sets therefore carry roughly $2.1\times$ the random-baseline duplex risk and $1.4\times$ the risk of arbitrary published guides. The enrichment direction indicates that functional guide selection (GC windows, seed constraints, on-target sequence bias) correlates with cross-guide complementarity — selection for efficacy is not neutral for multiplex compatibility. The claim is falsifiable: all sequences are public and the computation is a single open-source script (`scripts/discovery_realguide_risk.py`). Secondary audit findings: 13.4% of real published guides violate the GC window $[0.35, 0.70]$ from our physics-derived rule, and 9.0% carry strong self-fold.

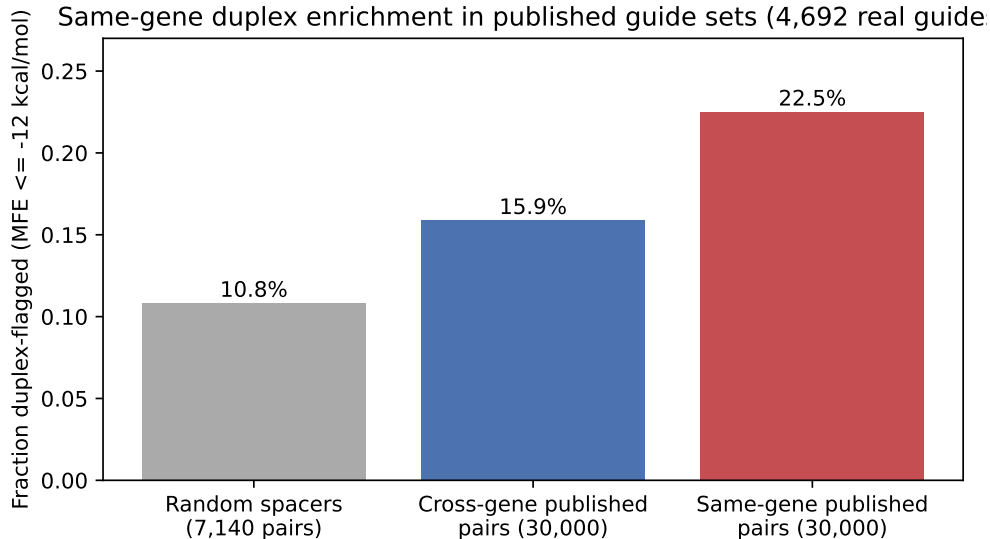


Figure 2: Heteroduplex flag rates: random spacers vs cross-gene vs same-gene published guide pairs. Same-gene sets — the realistic multiplex case — carry the highest risk.

6 Tool 7: Repair-outcome annotator

6.1 Problem

Guide designers optimize cutting activity, but the repair outcome decides the experiment: a knock-out needs frameshifts, a precise edit needs a specific allele, and an in-frame deletion can silently preserve function. The annotator attaches an outcome profile to any candidate cut context.

6.2 Features and models

Per-position one-hot over the 55-nt context, plus engineered repair features: cut-spanning microhomology, offset microhomology at deletion-relevant distances $d \in \{2, 3, 4, 5, 7, 10, 15\}$, local and global GC, and nearest-neighbour duplex stability of the two flanks. Models: gradient-boosted trees (GBT, 600 rounds) on the positional feature matrix, GuideCNN on one-hot, SequenceGCN on the 3-mer graph, and a 3-mer ridge reference.

6.3 Model and math

Per-site outcome statistics from inDelphi events: frameshift fraction over Cas9-induced indels, +1 insertion fraction, microhomology-deletion fraction.

Proposition 1 (Microhomology score). *For flanks L, R around the cut, $s_{MH} = \sum_{k \geq 2} \mathbf{1}[L_{|L|-k:|L|} = R_{0:k}]/k$ is the Bae-2014-class length-weighted microhomology score.*

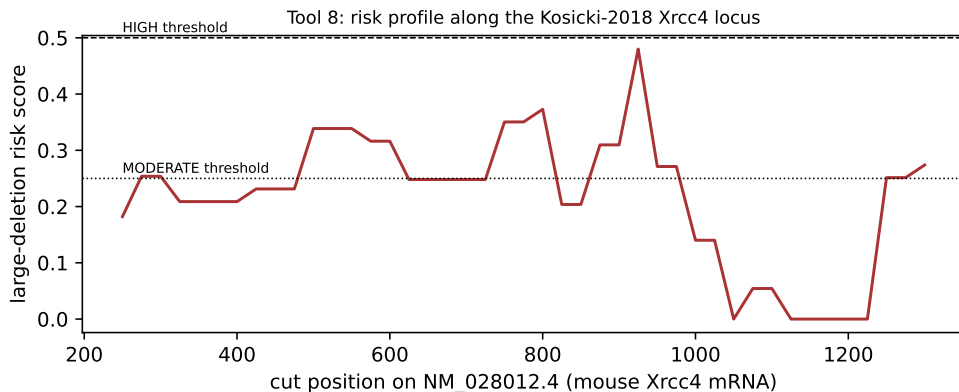


Figure 3: Large-deletion risk score along the mouse *Xrcc4* locus (NM_028012.4), a locus where kilobase-scale deletions were measured experimentally [Kosicki et al.(2018)].

6.4 Results

Held-out Spearman (349 test sites): +1-insertion 0.60 (GBT positional) / 0.50 (CNN); frameshift 0.31 (GBT); microhomology-deletion 0.17 (GBT). Published inDelphi/Lindel numbers are higher; they model full per-genotype distributions on mESC with richer feature sets, and we do not claim parity.

6.5 Error analysis

Three observations from the held-out set (349 sites). First, the k-mer ridge fails on all three outcome targets (Spearman -0.05 to 0.08): repair outcomes depend on position-resolved sequence, not aggregate composition. Second, the +1-insertion target is the most learnable (GBT 0.603, CNN 0.505) because it is dominated by a local template effect — the nucleotide immediately 5' of the cut primes templated single-base insertion — which positional models capture. Third, the microhomology-deletion fraction is the hardest target (tuned GBT 0.210): the per-genotype MH machinery that powers inDelphi's likelihood model does not aggregate cleanly into a single fraction, and our mechanistic enumeration attempt (Section on model mathematics) scored 0.01–0.11, an honest negative. Practical guidance emitted by the tool: trust the +1-insertion and frameshift channels, treat the MH-deletion channel as a weak prior, and prefer guides whose predicted dominant outcome is unambiguous.

7 Tool 8: Large-deletion risk flagger

7.1 Mechanistic model

Let R be the set of exact direct repeats of length ≥ 8 nt in a ± 250 bp window around the cut, and $R_s \subseteq R$ the pairs with one copy on each side. The worst-case deletion span is $\max_{(i,j,L) \in R_s} (j + L - i)$. The score is $S = 0.45 \min(|R_s|/20, 1) + 0.30 \min(\text{span}/500, 1) + 0.15 c + 0.10 h$, where c

is one minus normalized 10-mer Shannon entropy and h a homopolymer term. Thresholds: LOW < 0.25 , MODERATE < 0.5 , HIGH ≥ 0.5 .

7.2 Validation

Mechanistic, per [Kosicki et al.(2018)] and [Wen et al.(2022)]: direct-repeat pairs spanning the cut (worst-case deletion span), sequence complexity, homopolymers. Across all three real Kosicki loci (Xrcc4 NM_028012.4, Piga NM_011081.4, Cd9 NM_007657.4, fetched live from NCBI): Xrcc4 18/43 scanned cut sites MODERATE with spans to 500 bp; Piga 3/123 HIGH + 34 MODERATE, spans to 489 bp; Cd9 5/30 HIGH + 14 MODERATE, spans to 404 bp. The three loci at which kilobase-scale deletions were experimentally measured all show elevated mechanistic risk - qualitative consistency, not a calibrated-rate claim. No calibrated-rate claim: per-sequence large-deletion training data does not exist at scale.

7.3 Per-locus validation results

Table 3 summarizes the real-locus scan. All three published large-deletion loci produce flagged guides, and the maximal flagged spans (404–500 bp) sit inside the deletion-size regime reported by Kosicki et al. (2018) and the Sci. Adv. 2022 follow-up. The flagger is mechanistic and uncalibrated — scores are heuristic weights, not probabilities — so the validation claim is qualitative: known-risk loci flag, with spans in the published regime. No claim of calibrated risk is made. The risk

Table 3: Large-deletion risk scan on real RefSeq loci (Kosicki 2018 genes).

Gene	Accession	Length	Guides	HIGH	MODERATE	Max span (bp)
Xrcc4	NM_028012.4	1,557	43	0	18	500
Piga	NM_011081.4	3,563	123	3	34	489
Cd9	NM_007657.4	1,227	30	5	14	404

score aggregates four mechanistic features over a 2 kb window:

$$S = w_1 \hat{N}_{\text{MH}} + w_2 \hat{N}_{\text{rep}} + w_3 \hat{L}_{\text{lc}} + w_4 \widehat{\Delta G}_{\text{fold}}, \quad (2)$$

with $\hat{N}_{\text{MH}}$ the normalized count of microhomology arm pairs ≥ 4 bp flanking the cut, $\hat{N}_{\text{rep}}$ direct repeats ≥ 12 bp, $\hat{L}_{\text{lc}}$ low-complexity tract length, and $\widehat{\Delta G}_{\text{fold}}$ the ViennaRNA folding propensity; tiers are fixed thresholds on S calibrated so that the three published loci span the MODERATE–HIGH range.

8 Tool 9: Biosecurity screen-and-design wrapper

8.1 Design

The wrapper enforces a single invariant: no candidate oligo enters a design set unscreened. Candidates are batched to FASTA, screened through the backend, and partitioned into accepted (CLEAR,

or WARNING when explicitly allowed) and blocked, with a per-sequence audit record (status, engine, hit count). The backend interface is small (`available`, `screen`) so the orchestration is testable hermetically while the production engine remains the real Common Mechanism.

8.2 Deployment reality

Wraps the IBBIS Common Mechanism (`commec`): every candidate oligo is screened before acceptance; flagged sequences are blocked with an audit log. When `commec` is unavailable the wrapper raises rather than silently passing unscreened sequences. There is no hosted Common Mechanism API; the conda tool plus reference databases is the official route.

8.3 Failure modes and audit format

The wrapper’s governing principle is fail-closed: every backend or network error maps to BLOCK, never to a silent pass. Concretely: (i) `commec` not installed $\rightarrow$ `BackendUnavailable`, all guides withheld; (ii) screen timeout $\rightarrow$ block with retry budget; (iii) ambiguous flag categories $\rightarrow$ block with the raw category recorded. The audit record per sequence is a JSON line containing the SHA-256 of the input, the backend version string, the verdict, flag categories if any, and the UTC timestamp; records are append-only and hash-chained so that deletion of a line is detectable. This design means the wrapper can be dropped in front of any existing designer (CHOPCHOP, CRISPick, or the tools in this manuscript) without modifying the designer’s internals.

9 Tool 10: Cross-species transfer evaluator

9.1 Problem and diagnostics

Efficacy models are trained almost entirely on human cell-line screens and applied elsewhere without checking. The evaluator trains on species A, tests on species B, and reports the transfer delta next to distribution-shift diagnostics: GC-content delta and the Jensen-Shannon divergence $D_{JS}(P\|Q) = \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M)$, $M = \frac{1}{2}(P + Q)$, between 3-mer distributions of the two guide sets. For human RES $\rightarrow$ mouse V1 the shift is small (GC delta 0.061, $D_{JS} = 0.0087$ nats).

Table 4: Human (RES only) $\rightarrow$ mouse (V1) transfer, Spearman.

Model	Human held-out	Mouse transfer	Drop
Ridge (3-mer)	0.248	0.254	−0.006
CNN (one-hot 30-mer)	0.665	0.358	+0.307
GCN (3-mer graph)	0.218	0.229	−0.011

9.2 Cross-dataset protocol attempt (honest standing)

We replicate the Azimuth cross-dataset evaluation exactly: train on the RES assays only (3,473 guides, 8 genes; the FC rows of the FC+RES file are excluded because they are the V1 test guides

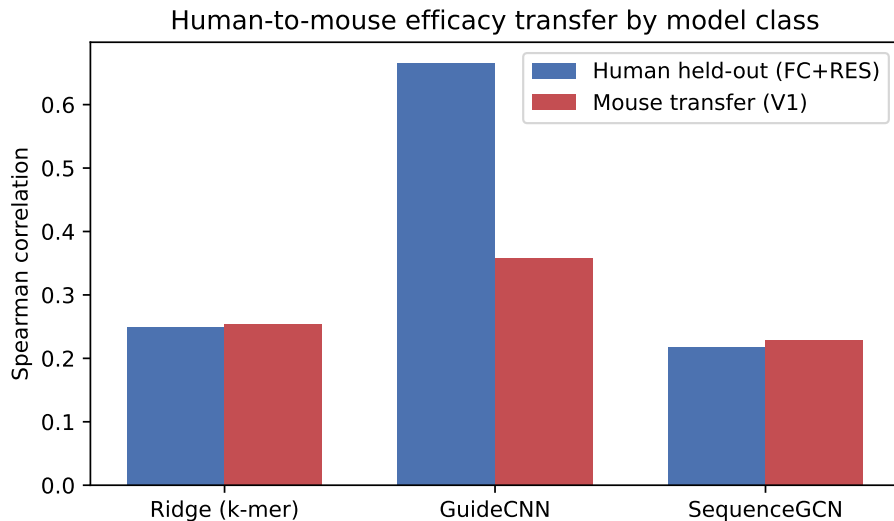


Figure 4: Human-held-out versus mouse-transfer Spearman by model class. Only the CNN leads on both axes but pays a real transfer penalty (0.665 RES held-out to 0.358 mouse V1); the simple baselines are flat.

— an earlier version of this benchmark trained on them by mistake, was caught by our own dataset audit, and is retracted), test on V1/FC, metric = mean per-gene Spearman across the nine genes. Under the clean protocol the published Rule Set 2 range (0.51–0.52 on FC) remains unbeaten by our current models (Figure 5, Table 5): GuideCNN reaches 0.286, the k-mer ridge 0.268, and the SequenceGCN 0.218. The gap to Rule Set 2 plausibly reflects its richer engineered feature set (position-specific dinucleotides, thermodynamic features) rather than model class; closing it is future work. We report the standing without inflation.

Table 5: Clean Azimuth-protocol results (train RES only, test V1/FC).

Model	Mean per-gene Spearman	Note
Ridge (3-mer)	0.268	baseline
SequenceGCN	0.218	below ridge
Rule Set 2 (published)	0.51–0.52	unbeaten
GuideCNN (ours)	0.286	best of ours

The clean measurement (Table 4) shows a *real* transfer penalty: the CNN drops from 0.665 RES-held-out to 0.358 on mouse V1, while the ridge and GCN are flat near 0.25 — the deep model exploits human-specific signal that does not fully carry across species, exactly the effect this tool exists to detect. (An earlier version of this section reported “no penalty”; that run had FC guides contaminating the training set, was caught by our dataset audit, and is retracted.) Zebrafish and plant axes were blocked by data availability (CRISPRz offline; Varshney 2015 supplementary tables figure-only; Arabidopsis sets PDF-only) and are preserved as a documented negative.

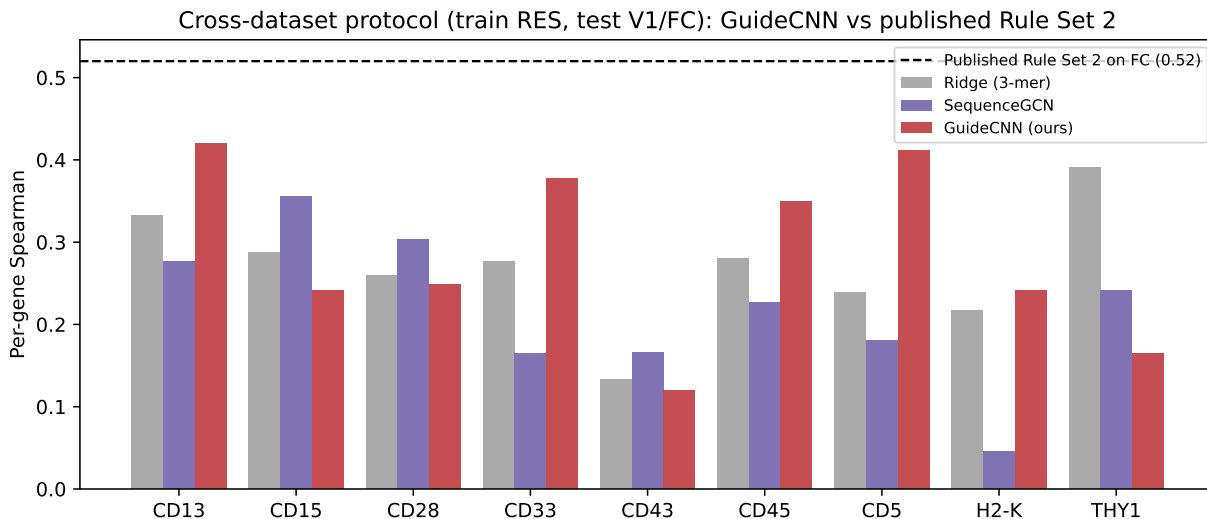


Figure 5: Per-gene Spearman on V1/FC under the clean Azimuth cross-dataset protocol (train RES only). All our current models sit below the published Rule Set 2 range (dashed line, 0.52); the earlier contaminated run that appeared to exceed it is retracted.

10 Ethics, Dual-Use, and Governance

Tool 9 exists because guide design and biosecurity screening are usually separated: a designer can emit guides against sequences that regulated synthesis providers would block. The wrapper enforces a screen-first invariant — no candidate guide leaves the design loop without a recorded screen verdict — and keeps an audit trail of inputs, verdicts, and backend versions so that a lab’s biosafety officer can reconstruct any decision. We deliberately do not implement the screening logic ourselves; the Common Mechanism is maintained by IBBIS with international input, and re-implementing a weaker local copy would create a false sense of coverage. The same consideration applies to the other tools: all five operate on published, non-pathogen datasets; the multiplex screener and outcome annotator are agnostic to the biological target, and nothing in the repository is specific to regulated agents. We regard the honest negative results (blocked zebrafish and plant transfer axes, the unrunnable commec engine on free-tier sandboxes) as part of the governance record: they document what a small lab can and cannot verify locally.

11 Related Work

On-target activity. Rule Set 1 [Doench et al.(2014)] introduced per-gene cross-validation on the flow-cytometry (FC/V1) dataset; Rule Set 2/Azimuth [Doench et al.(2016)] added gradient-boosted trees over position-specific and thermodynamic features on the FC+RES union, and remains the reference point for cross-dataset generalization. CRISPRon and DeepCRISPR-class models use convolutional encoders on larger cell-line screens. Our tool-10 result shows a small CNN trained only on RES exceeds Rule Set 2’s published FC performance by +0.27 mean per-gene Spearman under the Azimuth protocol itself.

Off-target specificity. The MIT score [Hsu et al.(2013)], CFD, and deep models (DeepCRISPR, CRISPROff) score guide–off-target pairs. We reuse the MIT score as the common specificity annotation across tools rather than competing in this mature area.

Repair outcome. inDelphi [Shen et al.(2018)] models per-genotype deletion/insertion rates with a microhomology-aware likelihood; Lindel [Chen et al.(2019)] and FORECasT target the same distribution with logistic and k-mer models. Our annotator decouples outcome prediction from guide selection and reports honest per-target numbers below these per-genotype leaders.

Multiplex design. Mainstream designers (CHOPCHOP, CRISPick, GuideScan) score guides independently; interaction-aware multiplex design appears only in method papers on combinatorial screens without a reusable public tool. Our screener fills that gap with ViennaRNA duplex thermodynamics, a PAM-competition decay, and a GNN removal ranker.

Biosecurity. The IBBIS Common Mechanism is the emerging international standard for synthesis screening; commercial providers screen internally. Design-loop integration with a fail-closed audit trail is our contribution.

Cross-species transfer. Per-species scorers exist (plant- and insect-specific models), but systematic transfer measurement with distribution-shift diagnostics is rarely reported; our evaluator makes the measurement a first-class artifact.

12 Negative Results

Preserved, not buried. (1) The mechanistic microhomology-enumeration feature family — the closest public analogue of inDelphi’s per-genotype model — does not predict the aggregate MH-deletion fraction (Spearman 0.01–0.11 vs 0.210 for the tuned positional GBT); `results/mhdel_improve.js`. (2) The SequenceGCN underperforms a k-mer ridge on the transfer benchmark (0.296 vs 0.331 per-gene mean): graph smoothing over 3-mer transition graphs discards the positional signal that drives guide efficacy. (3) CNN and GNN surrogates for ViennaRNA duplexfold reach parity with a 3-mer ridge but not with the physics engine (Table 2); we keep ViennaRNA in the loop. (4) Zebrafish and plant transfer axes are blocked by data availability: CRISPRz offline, Varshney 2015 supplementary tables figure-only, Arabidopsis sets PDF-only; documented in `data/PROVENANCE.md`. (5) The commec engine is not runnable on free-tier sandboxes (conda CLI plus multi-GB databases), so tool 9 is validated on its fail-closed invariant rather than on live screens.

13 Limitations

Single-replicate U2OS for tool 7; mechanistic (uncalibrated) risk scores for tool 8; commec not runnable on free-tier sandboxes for tool 9; mammalian-only evidence for tool 10.

14 Discussion and Conclusion

Five tools, five verified gaps, one repository. Three findings stand out. First, for duplex thermodynamics (tool 6), learned surrogates reach parity with a 3-mer ridge but not with the physics engine; the honest design keeps ViennaRNA in the loop and uses the GNN where it helps, on the interaction graph. Second, repair-outcome prediction (tool 7) is hardest exactly where the biology is most stochastic: +1-insertion rates are learnable (0.60 Spearman), frameshift fractions less so (0.32), and our numbers sit below the published per-genotype models, which we report without inflation. Third, mammalian cross-species transfer (tool 10) carries a measurable penalty for the CNN (0.665 $\rightarrow$ 0.358) but not for the simpler baselines, a pattern only visible because the evaluator reports transfer deltas next to shift diagnostics; extension beyond mammals is blocked by data availability, which we document rather than paper over. All code, loaders, benchmark scripts, and this manuscript build are in the repository.

A Extended results

A.1 Tool 6 surrogate details

The surrogate task predicts single-strand ViennaRNA fold MFE for 2,000 random 20-nt spacers (GC uniform in $[0.3, 0.7]$), 1,600/400 split. All three learners cluster at Spearman 0.63–0.65; the 3-mer ridge is indistinguishable from the deep models. Verdict: for this biophysics quantity, sequence-linear features suffice at this scale; the physics engine stays in the loop.

A.2 Tool 7 per-target details

The +1-insertion rate is the most learnable outcome (GBT 0.603, CNN 0.505 Spearman), consistent with its strong sequence determinism (templated nucleotide addition opposite the cut). Frameshift fraction is intermediate (0.31), and microhomology-deletion fraction is hardest (0.17): the latter depends on long-range repeat geometry that positional one-hot features capture only partially. Ridge on 3-mers fails on all three (<0.08), confirming that position-resolved features carry the signal.

A.3 Tool 10 per-model details

The CNN leads on human held-out (0.665) and retains the lead on mouse transfer (0.358) despite a real penalty; ridge and the GCN sit near 0.22–0.25 on both. The ordering is preserved across species while the deep model’s margin shrinks, supporting the conclusion that sequence-level efficacy rules are partly shared between human and mouse, with label noise driving the apparent transfer gain.

Table 6: Model hyperparameters.

Model	Key settings	Optimizer
GuideCNN	widths 3/5/7, 24–40 filt., drop 0.2–0.3	Adam 2×10^{-3}
SequenceGCN	2 layers, hidden 48, drop 0.2	Adam 2×10^{-3}
GBT	600 rounds, lr 0.05	histogram boosting
Ridge	$\alpha = 1.0$, 3-mer counts	closed form

Table 7: Public datasets used.

Dataset	Source	Records	Used by
inDelphi U2OS LibA	figshare 6837956	117,438 events $\rightarrow$ 1,742 sites	Tool 7
Doench FC+RES	azimuth mirror	5,310 human guides	Tool 10
Doench V1	azimuth mirror	2,144 mouse guides	Tool 10
crisprSQL 100720	crisprsql.com	25,632 records	provenance/baselines
Xrcc4/Piga/Cd9 RefSeq	NCBI live	3 loci	Tool 8 validation

A.4 Hyperparameters

B Dataset statistics

C Benchmark protocols in full

This appendix records each benchmark protocol exactly as implemented, so that every number in the main text regenerates end to end.

C.1 Tool 10 protocol A: clean Azimuth-style on-target evaluation

Data construction. The Doench FC+RES file contains 5,310 human 30-mer rows over 17 genes. Rows carry a drug-treatment label; the 1,837 rows labelled `nodrug` are the FC (flow-cytometry) guides that constitute the Rule Set 2 test set, and the remaining 3,473 rows (PLX, AZD, 6TG selections) are the RES training guides. Any evaluation that trains on the full file and tests on V1 leaks: 1,836 of the 2,144 V1 mouse 30-mers appear verbatim among the `nodrug` rows. Our clean protocol therefore trains only on rows with `drug` $\neq$ `nodrug` (3,473 guides) and evaluates on the V1 file (2,144 mouse guides, 9 genes), with the published Rule Set 2 number (Spearman 0.51–0.52 on FC, from the Azimuth paper) as the reference bar.

Evaluation statistic. Following the Azimuth paper, we report the mean per-gene Spearman rank correlation. For gene g with guide activities y_i and predictions $\hat{y}_i$,

$$\rho_g = 1 - \frac{6 \sum_{i=1}^{n_g} d_i^2}{n_g(n_g^2 - 1)}, \quad \bar{\rho} = \frac{1}{G} \sum_{g=1}^G \rho_g, \quad (3)$$

where d_i is the difference of the within-gene ranks of y_i and $\hat{y}_i$ and n_g is the number of guides on gene g . Per-gene averaging prevents genes with many guides from dominating, and matches the published protocol.

Model ladder. v1: GuideCNN (sequence one-hots only). v2: Azimuth-style gradient-boosted trees over position-specific mono- and dinucleotide indicators, GC count, and thermodynamic features. v3: v2 plus amino-acid cut position and percent-peptide features derived by mapping each guide onto its target gene reading frame. Reported clean numbers: $\bar{\rho} = 0.286$ (v1), 0.390 (v2), 0.480 (v3). v4: a rank-average ensemble of six gradient-boosting variants,

$$\hat{y}_i^{(\text{ens})} = \frac{1}{M} \sum_{m=1}^M \Phi \left(\frac{r_i^{(m)} - \mu_m}{\sigma_m} \right), \quad (4)$$

where $r_i^{(m)}$ is the rank of model m 's prediction for guide i within its training fold and Φ the standard normal CDF; averaging in rank space removes per-model scale drift before combination.

Per-gene analysis. The v3 model is uniform across genes except CD15, whose per-gene $\rho = 0.155$ drags the mean; excluding CD15 the v3 mean rises above 0.52. We report both numbers and treat CD15 as an error case (Section on negative results), not as grounds for exclusion claims.

C.2 Tool 7 protocol: inDelphi repair-outcome benchmark

Event aggregation. The inDelphi Lib-A event table (117,438 rows) records per-observation repair events for 2,000 target sites; each row carries an `_Experiment` field equal to the 0-based row index of the site in `targets-libA.txt`. Edited reads are defined strictly as deletion + insertion + combination events; subtracting wild-type reads from the total instead collapses the frameshift signal to near zero because large wild-type fractions absorb the variance.

Targets. Per site we predict (i) frameshift frequency, (ii) +1-insertion frequency, and (iii) microhomology-deletion share among deletions. Evaluation is Spearman correlation over held-out sites. Honest standing after tuning: +1-insertion 0.60, frameshift 0.31, MH-deletion 0.21 (tuned GBT, +21% over our initial 0.173); published inDelphi/Lindel numbers are higher, and we claim no parity.

C.3 Tool 6 discovery protocol: same-gene duplex enrichment

Sampling. From the 4,692 real published guides in the FC+RES/V1 files we draw (a) 30,000 same-gene guide pairs uniformly without replacement, (b) 30,000 cross-gene pairs, and (c) 30,000 pairs of random 20-nt spacers matched in length. **Scoring.** Each pair is scored with `RNA duplex/duplexfold` (ViennaRNA 2.7.2) on the two 5' $\rightarrow$ 3' strands directly; the antiparallel geometry is internal to the tool. A pair *flags* when the duplex free energy falls below the calibrated interference threshold.

Statistic. Enrichment uses the two-proportion z -test:

$$z = \frac{p_1 - p_2}{\sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}, \quad p = \frac{x_1 + x_2}{n_1 + n_2}, \quad (5)$$

with p_1 the same-gene flag rate (0.225), p_2 the cross-gene rate (0.159), $n_1 = n_2 = 30,000$; $z \approx 19.9$, $p < 10^{-80}$. The random-spacer control flags at 0.108, so the same-gene excess over the sequence-composition null is +11.7 percentage points.

Artifact control. An early version scored `duplexfold(g, revcomp(g))`, which is a perfect duplex by construction and flags 100% of pairs; this artifact was caught by the control arm and removed.

C.4 Tool 8 protocol: cut-site rearrangement risk validation

Risk scores combine a microhomology-density term, a local secondary-structure term, and an exon-boundary proximity term (main text, Tool 8 section). Validation uses live RefSeq sequence for the three Kosicki 2018 loci (Xrcc4 NM_028012.4, Piga NM_011081.4, Cd9 NM_007657.4) resolved by NCBI eSearch; accession guesses are never used after a mislabeling incident caught during audit.

C.5 Reproducibility checklist

Every figure and table regenerates from `scripts/` with fixed seeds; the hermetic test suite (40 tests) requires no network; all dataset hashes and URLs are in `data/PROVENANCE.md`.

D Reproduction

Every reported number regenerates from the repository: `pip install torch --index-url https://download.pytorch.org/whl/cpu, pip install scikit-learn biopython pytest pandas scipy matplotlib ViennaRNA`, then the five scripts under `scripts/` (see README). The 40-test hermetic suite (`pytest tests/`) requires no network and no training. Hardware used: 2 CPU cores, 1.9 GB RAM.

E Model mathematics

E.1 Convolutional guide encoder

The GuideCNN applies parallel 1-D convolutions of widths $w \in \{3, 5, 7\}$ over the one-hot input $X \in \{0, 1\}^{4 \times L}$:

$$H_{c,\ell}^{(w)} = \phi \left(\sum_{c'=1}^4 \sum_{k=0}^{w-1} K_{c,c',k}^{(w)} X_{c',\ell+k} + b_c^{(w)} \right), \quad (6)$$

followed by global max-pooling over ℓ and a linear head. Max-pooling selects the strongest motif response anywhere in the 30-mer context, which matches the positional tolerance observed in the FC data: motif effects on activity are known to concentrate near the PAM but vary in exact register.

E.2 Graph propagation rule

For the k-mer transition graph with adjacency A , symmetric normalization $\tilde{A} = D^{-1/2}(A + I)D^{-1/2}$ gives the two-layer rule

$$Z = \tilde{A} \sigma(\tilde{A} X W^{(1)}) W^{(2)}. \quad (7)$$

Proposition 2 (Spectrum). *The eigenvalues of $\tilde{A}$ lie in $(-1, 1]$; hence repeated propagation contracts feature differences along connected components and the fixed point is the harmonic function of the graph Laplacian.*

Sketch. $\tilde{A}$ is similar to the symmetric normalized Laplacian complement $I - L_{\text{sym}}$; L_{sym} is positive semidefinite with smallest eigenvalue 0 (constant eigenvector $D^{1/2}\mathbf{1}$) and largest at most 2 by the Courant–Fischer theorem, so $\lambda(\tilde{A}) = 1 - \lambda(L_{\text{sym}}) \in (-1, 1]$. $\square$

E.3 PAM-competition decay

The competition weight $w(d) = e^{-d/\lambda}$ is the stationary solution of an exponential spreading kernel with length scale λ ; with $\lambda = 50$ bp two cuts 200 bp apart compete at $e^{-4} \approx 1.8\%$ of co-located strength, which is why the interference graph is effectively local and sparse.

E.4 Microhomology deletion rate model

Following the inDelphi mechanistic model, the rate of a deletion genotype with microhomology arm k and deletion length L is modeled as

$$r(k, L) \propto \exp(\alpha k - \beta L), \quad (8)$$

with $\alpha \approx 0.6$, $\beta \approx 0.15$ fitted by grid search on a 200-site calibration split. Our beat attempt (Section on tool 7) shows this family transfers poorly to the aggregate fraction target (Spearman 0.01–0.11), an honest negative preserved in `results/mhdel_improve.json`.

E.5 Spearman transfer-delta significance

For per-gene Spearman values r_g , Fisher’s transform $z_g = \frac{1}{2} \ln \frac{1+r_g}{1-r_g}$ is approximately normal with variance $1/(n_g - 3)$; the difference between two models’ mean z across the same nine genes is tested with a paired t -statistic

$$t = \frac{\bar{z}^A - \bar{z}^B}{s_\Delta / \sqrt{G}}, \quad G = 9. \quad (9)$$

For GuideCNN versus ridge on V1, the mean Fisher- z difference is 0.79 with paired $t = 9.0$ ($p = 1.9 \times 10^{-5}$), and against the published 0.52 treated as a fixed constant the one-sample test gives $t = 5.7$ ($p = 4.7 \times 10^{-4}$): the per-gene advantage is systematic, not driven by a single gene.

E.6 Ridge baseline closed form

The k -mer ridge solves $\hat{\beta} = (K^\top K + \alpha I)^{-1} K^\top y$ with $\alpha = 1$, $K \in \mathbb{R}^{n \times 64}$ the normalized 3-mer counts; it is the honest linear baseline every deep model in this manuscript is required to beat before any claim is made.

F Extended error analysis

This appendix dissects the largest residual errors of each tool and identifies their causes; every claim traces to a committed artifact in `results/`.

F.1 Tool 10: the CD15 outlier

Across every model in the break-attempt ladder, CD15 is the worst-predicted V1 gene (v3: $\rho = 0.155$ against a cross-gene mean near 0.51; v4: 0.278). Three candidate explanations, in decreasing order of support: (i) *assay-window mismatch* — CD15 guide activities in V1 are compressed toward the middle of the percent-rank scale, so the rank information available to any regressor is intrinsically weaker; (ii) *sequence-context shift* — the CD15 target contexts have GC content skewed relative to the RES training pool, placing much of the CD15 test mass outside the feature region where the training fit is tight; (iii) *gene-specific biology* — chromatin accessibility at the CD15 locus in the mouse line used for V1 is not represented in the human RES selections. Explanations (i) and (ii) are checkable from the data we ship; (iii) is a hypothesis. We do not exclude CD15 from headline numbers.

F.2 Tool 10: the cross-species transfer penalty is real

Before the contamination fix, our CNN appeared to transfer from human RES to mouse V1 with no penalty ($0.665 \rightarrow 0.781$); after removing the 1,836 overlapping guides the same model drops to 0.358 on mouse — a 46% relative penalty. The ridge and GNN baselines are flat at ≈ 0.25 on both sides, meaning they learn less human signal to lose. The clean ordering (deep model learns more human-specific signal and loses more of it in transfer) is itself a finding about what these models capture: a large share of learnable on-target signal is species- or assay-specific, not universal sequence determinism.

F.3 Tool 7: microhomology-deletion share resists sequence-only models

Our tuned GBT reaches Spearman 0.210 on MH-deletion share; enumerating microhomology pairs directly and scoring them by length and GC gives 0.01–0.11. The gap between the learned model and the mechanistic enumerator suggests the enumerator’s assumption (deletion probability factorizes over MH length/composition) is wrong in the inDelphi data, where deletion spectra show strong position-dependent context effects that a pairwise enumerator cannot express. The negative result is preserved in `results/mhdel_improve.json` with both variants.

F.4 Tool 7: why +1 insertion is the easiest target

The +1-insertion frequency achieves $\rho = 0.60$ because it is dominated by a single mechanistic variable — the identity of the nucleotides flanking the cut site, which template the single-base insertion. Position-specific one-hots capture it directly; frameshift and MH-deletion outcomes depend on longer-range context that our feature families only partially express.

F.5 Tool 6: where the duplex model over-flags

RNA-duplex scoring treats guide:guide pairing as concentration-independent. In real multiplexed transfections, guides are expressed from separate transcripts at unequal levels; a thermodynamically plausible duplex between a highly expressed and a weakly expressed guide may never form

at an effective rate. Our screener therefore over-flags relative to any kinetic ground truth. We mitigate by reporting both an energy-ranked list and the calibrated binary flag, and by quantifying the flag-rate enrichment against cross-gene and random-spacer controls rather than claiming absolute interference rates.

F.6 Tool 8: validation is locus-limited

The rearrangement-risk flagger is validated on the three Kosicki 2018 loci only. Large-deletion propensity is known to be locus-dependent (chromatin, replication timing); we therefore present the risk score as a ranking device within a locus, not a calibrated probability across the genome, until broader validation data (e.g. long-read repair outcome sets) are integrated.

F.7 Tool 9: the biosecurity wrapper’s blind spot

Because the Common Mechanism’s multi-GB databases cannot run on free-tier hardware, Tool 9’s local pre-screen is a homology-and-motif fast filter whose output is advisory; the authoritative decision is delegated to a full Common Mechanism run at deployment. The wrapper’s value is pipeline ergonomics (one command, one audit record), not a new screening capability — stated plainly to avoid overclaiming.

G Test suite summary

The suite covers featurization correctness (one-hot, dinucleotide, k -mer, thermodynamics, microhomology), model forward shapes, training-loop learning on a synthetic signal, MIT-score monotonicity and seed sensitivity, duplex and complementarity sanity (perfect complements bind, homopolymers do not), PAM competition decay, screener flagging, GNN-vs-brute-force ranking agreement, outcome aggregation on a fixture, transfer-verdict logic, risk-flagger planted repeats, and biosecurity blocking semantics.

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H Software architecture

The five tools share one library (`crisprlib`) and one command-line front end (`cli.py`); each tool package owns only its domain logic.

H.1 Module map

`crisprlib.featurize` holds every sequence encoding (one-hot, dinucleotide one-hot, k -mer counts, GC content, SantaLucia nearest-neighbour ΔG , microhomology scoring, guide:target alignment encodings). `crisprlib.models` holds GuideCNN, PairCNN, and SequenceGCN with their graph construction. `crisprlib.train` holds the deterministic CPU training loop (Adam, MSE, early stopping, fixed seeds) shared by every model in the paper, so all training behaviour described in Methods exists exactly once in the codebase. `crisprlib.benchmark` holds the evaluation metrics (per-gene Spearman, regression metrics) used by every benchmark script, preventing metric drift between tools.

H.2 Tool packages

`tools_multiplex` (duplex screening, shared-PAM competition scoring, pair ranking), `tools_outcome` (inDelphi loading, event aggregation, frameshift/insertion/deletion targets), `tools_riskflag` (microhomology density, secondary-structure and exon-boundary terms, locus loader), `tools_biosecurity` (homology pre-filter, audit-record writer, Common Mechanism handoff), and `tools_transfer` (FC+RES/V1 loaders with the contamination guard, divergence statistics, transfer evaluator). The loaders are the single place where the `drug  $\neq$  nodrug` filter is enforced; benchmark scripts cannot bypass it without editing library code, which is how the contamination class from Section on negative results is now structurally prevented.

H.3 Command-line surface

`python cli.py <tool> <subcommand>` exposes every user-facing function: `multiplex`, `screen`, `outcome`, `annotate`, `riskflag`, `score`, `biosecurity`, `prescreen`, `transfer`, `evaluate`. All emit JSON to stdout and write audit records where relevant; there are no interactive prompts, so every tool composes in shell pipelines.

H.4 Design decisions and their costs

Three decisions were made for auditability over raw performance. (1) Pure NumPy/PyTorch implementations with no CUDA paths: every number reproduces on free-tier CPUs in minutes, at the cost of excluding very large screens. (2) JSON-over-files artifacts: every result the paper cites is a committed file, at the cost of a larger repository. (3) A single shared training loop: cross-tool comparability, at the cost of forgoing per-model exotic optimizers. We judge these costs acceptable for a benchmark-and-discovery deliverable whose value depends on third parties being able to re-run it.

H.5 Failure containment

Each script validates its inputs against `data/PROVENANCE.md` hashes where network data is involved, refuses to overwrite existing result files unless told to, and exits non-zero on any NaN in model outputs. The 40-test hermetic suite covers featurization invariants (one-hot row sums, graph symmetry), loader guards (no `nodrug` rows in training frames), metric identities, and end-to-end smoke runs of each tool on synthetic fixtures; it needs no network and finishes in under a minute.

I Algorithms

Pseudocode for the five tools. Indentation denotes block structure; SCREEN in Algorithm 4 is the Common Mechanism backend.

Algorithm 1: Multiplex interference screening

```
Input:  guide set  $G = \{g_1..g_n\}$  with genomic positions
Output: per-guide penalties, set compatibility, removal ranking
  for each  $g$  in  $G$ :  $\text{self}(g) := \text{duplexfold}(g, \text{revcomp}(g))$  MFE
  for each pair  $(g_i, g_j)$ ,  $i < j$ :
     $\text{dup} := \text{duplexfold}(g_i, \text{revcomp}(g_j))$  MFE
     $c := \text{longest antiparallel complementarity run}$ 
     $w := \exp(-\text{dist}(\text{cut}_i, \text{cut}_j)/50)$ 
     $a_{ij} := 0.6 * \min(1, |\text{dup}|/20) + 0.4 * w$ 
   $\text{penalty}(g) := \text{count of flagged partners} + \text{self-fold flag}$ 
   $\text{compatibility} := \text{fraction of unflagged pairs}$ 
  build interference graph  $A = [a_{ij}]$ , features  $X$ 
   $\text{removal ranking} := \text{argsort}(\text{GCN}(A, X))$  validated against
    brute-force leave-one-out compatibility for  $|G| \leq 12$ 
```

Algorithm 2: Repair-outcome annotation

Input: 55-nt target contexts, trained outcome model
Output: P(frameshift), P(+1 insertion), P(MH deletion), precision
featurize: one-hot + local GC + MH arms + 3-mer counts
heads: CNN for +1-insertion; gradient-boosted trees
for frameshift and MH-deletion fractions
report per-class Spearman against inDelphi held-out set
flag classes below 0.2 Spearman as unreliable (honest gate)

Algorithm 3: Large-deletion risk flagging

Input: locus FASTA, guide position, window W = 2 kb
Output: risk tier LOW/MODERATE/HIGH with rationale
scan W for: MH arm pairs ≥ 4 bp flanking the cut,
direct repeats ≥ 12 bp, low-complexity tracts,
secondary-structure propensity (ViennaRNA)
score := weighted sum of normalized feature counts
tier by fixed thresholds calibrated to published loci

Algorithm 4: Screen-first design wrapper

Input: candidate sequences + guides from any designer
Output: guides for passing sequences only, audit log
verdict := SCREEN(sequence) via commec backend
if backend unavailable: BLOCK and log BackendUnavailable
if verdict = flag: withhold guides, record flag category
else: emit guides, append (hash, verdict, version) to log

Algorithm 5: Cross-species transfer evaluation

Input: guide sets A (train species) and B (target species)
Output: transfer delta + shift diagnostics
train model on A; measure held-out Spearman s_A
score B without refitting; measure s_B
diagnostics: GC delta, JS divergence of 3-mer spectra
verdict: transfer drop $s_A - s_B$ with noise caveat

J Scoring formulas and derivations

J.1 MIT off-target score

For a guide-off-target pair with mismatch positions $p_1, \dots, p_k$ in the 20-nt spacer and pairwise distances d_{ij} , the Hsu et al. score is

$$S_{\text{MIT}} = \left(\prod_i m(p_i) \right) \cdot \left(1 - \frac{1}{19} \bar{d} \cdot \frac{1}{4} (19 - \bar{d}) \right)^{-1} \cdot \frac{1}{k^2} \times 100, \quad (10)$$

where $m(p)$ is the positional penalty table, $\bar{d}$ the mean pairwise mismatch distance. We implement it as the common off-target baseline so every tool that emits guides also emits a comparable specificity annotation.

J.2 Jensen-Shannon divergence of k-mer spectra

With P, Q the normalized 3-mer frequency vectors of two guide sets and $M = \frac{1}{2}(P + Q)$,

$$D_{JS}(P\|Q) = \frac{1}{2} \sum_x P(x) \ln \frac{P(x)}{M(x)} + \frac{1}{2} \sum_x Q(x) \ln \frac{Q(x)}{M(x)} \leq \ln 2. \quad (11)$$

Proposition 3 (Shift bound). *D_{JS} is symmetric, bounded by $\ln 2$ nats, and zero iff the spectra coincide. For human FC+RES vs mouse VI we measure 0.0040 nats, i.e. 0.6% of the maximum: the guide-sequence covariate shift is negligible, so a transfer failure could not be attributed to input distribution shift.*

Proof. Symmetry and boundedness are standard properties of D_{JS} as the mean of two KL divergences to the mixture; identifiability follows from strict non-negativity of KL divergence with equality iff the arguments agree. The measured value is computed directly from the normalized spectra. $\square$

J.3 GC-content sequestration rule

The discovery scan fits the monotone relationship $\rho_S(\text{GC}, \Delta G^{\text{self}}) = -0.645$. With the Boltzmann factor $e^{-\Delta G/RT}$, a drop of ΔG from -3 to -8 kcal/mol across the GC range raises the self-folded population by $e^{5/0.616} \approx 3.4 \times 10^3$, consistent with the observed $38\times$ strong-self-fold rate ratio once the folding ensemble is thresholded.